DemandVerse AI: An Adaptive Retail Intelligence Framework for Demand Forecasting, Inventory Optimization, and Financial Impact Analysis

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***Abstract*—Retail supply chains operate under severe demand uncertainty, where misaligned inventory replenishment generates acute financial penalties through out-of-stock lost sales or excessive holding costs and margin-eroding markdowns. Existing forecasting systems frequently function in computational silos, disconnecting time-series predictions from buffer sizing, multi-scenario stress-testing, and balance-sheet risk quantification. This paper presents DemandVerse AI, an end-to-end retail intelligence framework that integrates statistical and additive time-series forecasting, automated model selection, dynamic inventory optimization, and scenario-based financial impact analysis. The framework was trained and empirically validated on 28,695 daily transaction records from the Walmart M5 benchmark across California retail operations (1,913 calendar days). DemandVerse AI implements parallel forecasting engines using Autoregressive Integrated Moving Average (ARIMA) and Meta Prophet models, governed by a hierarchical out-of-sample error routing mechanism evaluated across Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE). The adaptive routing mechanism selected Prophet for 11 stock-keeping units (73.3%) exhibiting pronounced calendar seasonality and ARIMA for 4 units (26.7%) governed by stationary autoregression, achieving an aggregate portfolio MAE of 5.0487, RMSE of 6.2916, and MAPE of 52.22%, outperforming isolated ARIMA and Prophet baselines across all metrics. The resulting 30-day forecast distributions dynamically parameterize a service-level safety stock ($Z = 1.645$) and Reorder Point (ROP) optimization engine. The framework represents the inventory state through an in-silico Digital Twin simulation sandbox, classifying the portfolio into 60.0% healthy stock, 26.7% overstock, and 13.3% stockout risk. Sensitivity stress-testing across three contractual demand shocks (-10%, Baseline, $+10\%$) establishes that procurement exposure shifts from 2,684.32 to 3,592.58 units, modulating stockout revenue-at-risk between $4,588.66 and $6,133.76, while holding cost carrying exposure ranges from $5.69 to $12.93 per month. The entire pipeline is deployed via an asynchronous FastAPI backend and an interactive React 19 executive dashboard, validated by 65 passing automated tests and a 24-point end-to-end verification harness.**

***Index Terms*—Retail Analytics, Demand Forecasting, ARIMA, Meta Prophet, Adaptive Forecasting, Inventory Optimization, Digital Twin, Time-Series Analysis, Financial Impact Analysis.**

I. Introduction

Modern omni-channel retail supply chains face unprecedented volatility driven by shifting consumer preferences, promotional events, seasonal variations, and macroeconomic disruptions [1]. Retail replenishment planners routinely grapple with two diametrically opposing operational failure modes: under-stocking and overstocking [1], [2]. Under-stocking precipitates stockouts, resulting in unrecoverable revenue loss, diminished consumer satisfaction, customer churn, and supplier contract penalties. Conversely, overstocking sequesters working capital in non-productive inventory, escalates warehouse storage, climate control, and insurance holding costs, and necessitates aggressive product markdowns or perishability write-offs [3]. Despite substantial advancements in predictive analytics, conventional enterprise resource planning (ERP) architectures frequently decouple the demand forecasting tier from operational inventory control and financial decision-making [4], [5].

In standard commercial workflows, statistical analysts generate unit-level demand forecasts in isolated tools, which are subsequently exported to static spreadsheets where replenishment managers manually apply heuristic rule-of-thumb order policies [5]. This fragmented workflow exhibits three critical structural deficiencies:

1) **Model Inflexibility:** Monolithic deployment of a single algorithm (e.g., global ARIMA or global Prophet) fails to capture the heterogeneous demand dynamics across

diverse retail categories, where some products exhibit autoregressive mean-reverting behavior while others display multi-frequency seasonal cycles [6], [7].

2) **Operational Disconnect:** Forecasted demand figures are rarely translated into rigorous, variance-aware buffer parameters, such as lead-time-adjusted safety stocks and dynamic Reorder Points (ROP) calibrated to defined cycle service levels [2], [8].
3) **Financial Blindness:** Forecast variances and inventory health states are infrequently translated into balance-sheet risk metrics, leaving executive stakeholders unaware of potential revenue-at-risk or capital drag until quarterly earnings disclosures [4].

To overcome these deficiencies, this research develops, implements, and evaluates **DemandVerse AI**, a unified retail intelligence framework. The framework transforms raw, multi-year transactional retail sales records into forward-looking demand trajectories, evaluates parallel time-series algorithms out-of-sample, dynamically routes each individual stock-keeping unit (SKU) to its optimal predictive engine, parameterizes a dynamic inventory control engine, simulates the inventory state via a Digital Twin sandbox, stress-tests supply chain resilience under demand shocks, and quantifies monetary exposure for corporate decision-makers. The primary contributions of this work are summarized as follows:

- **Dual-Engine Adaptive Forecasting:** We formulate and deploy a dual-engine architecture incorporating statistical Autoregressive Integrated Moving Average (ARIMA$(1,1,1)$) and additive Meta Prophet models, governed by a multi-tier tie-breaking model router that achieves superior out-of-sample predictive precision over single-model baselines.
- **Service-Level Inventory Intelligence:** We couple forward-looking forecast distributions with historical demand volatility (σ_d) and replenishment lead times ($L = 7$ days) to formulate dynamic Safety Stock (SS), Reorder Points (ROP), and automated order recommendations that classify SKU health states into Healthy Stock, Overstock, and Stockout Risk.
- **Digital Twin Inventory State Representation:** We model the inventory state through an in-silico simulation sandbox, establishing a clear academic and operational boundary between simulated inventory states and live IoT-instrumented physical warehouse twins.
- **Multi-Scenario Financial Risk Engine:** We formulate sensitivity stress-testing under -10%, baseline, and $+10\%$ demand shocks, quantifying projected gross revenues, stockout revenue-at-risk, and inventory holding costs using actual shelf prices from the Walmart M5 benchmark.
- **End-to-End Enterprise Deployment:** We deliver a containerized microservice architecture featuring an asynchronous FastAPI RESTful service (with LRU query caching) and a high-performance React 19 Single Page Application (SPA), verified via 65 automated software unit tests and a 24-point audit harness.

The remainder of this paper is organized as follows: Section II reviews relevant literature. Section III identifies specific research gaps. Section IV presents the formal problem formulation and project objectives. Section V and Section VI detail the proposed methodology, system architecture, dataset characteristics, and preprocessing pipelines. Section VII and Section VIII formulate the forecasting and inventory optimization mathematical engines. Section IX and Section X elaborate on the Digital Twin inventory state representation and what-if sensitivity analysis. Section XI defines the financial quantification models. Section XII details the software technology stack. Section XIII presents the experimental results and empirical discussion. Section XIV outlines study limitations. Finally, Section XV concludes the paper and delineates future research vectors.

II. Related Work and Literature Survey

Time-series forecasting and inventory replenishment have constituted a central pillar of operational research for over half a century [1], [3]. Box, Jenkins, Reinsel, and Ljung [9] formalized the classical Autoregressive Integrated Moving Average (ARIMA) methodology, providing a robust statistical foundation for linear stationary and non-stationary processes through lag polynomials and iterative differencing. While ARIMA remains an industry benchmark for short-term univariate forecasting, its parameter estimation requires strict variance stationarity and struggles to capture complex calendar anomalies and non-linear holiday shocks without extensive manual intervention [10].

To address calendar-rich time series at enterprise scale, Taylor and Letham [11] introduced Meta Prophet, an additive generalized additive model (GAM) that decomposes non-linear trends into piecewise logistic or linear growth, Fourier-series seasonal cycles, and irregular holiday regressors. By utilizing a Bayesian framing optimized via L-BFGS or MAP estimation, Prophet handles missing data, structural changepoints, and weekly-yearly retail seasonality with high robustness. Comparative retail forecasting studies by Salerno [6], Bandara, Bergmeir, and Smyl [12], Dogan [7], and Carbonneau, Laframboise, and Vahidov [13] underscored that while Prophet and machine learning models excel on series with prominent weekend and holiday pulses, classical autoregressive formulations frequently match or exceed complex models when the underlying demand exhibits localized mean-reverting autoregression. Multi-step forecasting evaluations by Taieb et al. [14], Nagalakshmi et al. [15], and intermittent neural benchmarks by Kourentzes [16] further confirm that model performance varies significantly across forecasting horizons and SKU velocity classes.

The landmark Walmart M5 Forecasting Accuracy competition organized by Makridakis, Spiliotis, and Assimakopoulos [17], [18] evaluated thousands of hierarchical retail forecasting pipelines across three states and ten stores. The results from the decision trees (GBDT) and neural networks were very good for the aggregate data but for the individual localized

SKU subsets, the more traditional statistical models such as the GBDTs and ARIMA models were superior in terms of parsimony, computational efficiency and even explainability. Spiliotis [4] further highlighted the imperative for dynamic, adaptive model selection frameworks that partition time series and dynamically route individual products based on localized error characteristics.

In inventory control, the seminal foundational theory established by Silver, Pyke, and Peterson [3] formulated continuous-review (s, Q) and periodic-review (R, S) replenishment policies based on the normal distribution of lead-time demand. Modern supply chain formulations by Chopra and Meindl [1] and Shubha Rao et al. [8] emphasize that static safety stocks derived solely from backward-looking variance induce bullwhip amplification when actual demand undergoes structural shifts. Coupling predictive demand trajectories directly with lead-time standard deviation enables dynamic buffer adjustment [2].

Recently, the concept of the Digital Supply Chain Twin has emerged as a major paradigm in Industry 4.0 [19]– [20]. Ivanov and Dolgui [19] define a digital supply chain twin as a computational representation of physical logistics networks that mirrors operational states and simulates the structural impact of disruptions. However, as noted by Wamba et al. [20], the term is frequently conflated in academic literature: researchers often label simulated inventory sandboxes as full real-time digital twins. In empirical retail research where real-time IoT feeds and warehouse RFID streams are absent (such as the Kaggle M5 benchmark), the digital twin operates legitimately as an in-silico state-space simulation that mirrors SKU stock transitions and evaluates operational stress without risking physical stockout exposure [21].

III. Research Gap Identification

A rigorous review of academic and industrial literature reveals five prominent research and operational gaps that DemandVerse AI directly addresses:

1) **Decoupling of Demand Prediction from Inventory Operations:** Prevailing research concentrates overwhelmingly on optimizing forecasting loss functions (e.g., minimizing RMSE) in complete isolation from downstream stock replenishment parameters [5], [18]. A mathematically superior forecast that is not translated into Safety Stock, Reorder Point, and purchase order quantities provides zero direct utility to procurement managers.
2) **Rigid Single-Model Bias:** Most enterprise implementations enforce a uniform model architecture across all SKUs, failing to recognize that retail portfolios contain heterogeneous demand profiles [6], [7]. Monolithic model assignment leads to severe sub-optimization when seasonal products are forced into linear autoregression or stationary products are overfitted with complex Fourier series.
3) **Static, Non-Adaptive Replenishment Thresholds:** Traditional ERP software calculates safety stock using annualized static variance, ignoring forward-looking demand shifts, promotional calendar triggers, and dynamic lead-time risk exposure [2].
4) **Absence of Pre-Shock What-If Sensitivity Testing:** Existing retail analytics solutions typically report historical stockouts retrospectively. Supply chain planners lack interactive simulation sandboxes to stress-test SKU stockout exposure and reorder requirements under hypothetical market shocks before physical replenishment decisions are finalized [21].
5) **Divergence Between Operational Units and Financial Exposure:** Academic forecasting papers report error metrics in abstract statistical terms (e.g., normalized MAE or RMSSE) without translating physical shortages or surplus into financial balance-sheet metrics (e.g., gross revenue-at-risk and capital carrying cost drag) [4].

DemandVerse AI bridges these gaps by delivering an integrated, mathematically rigorous, and fully functional intelligence pipeline that couples dual-model forecasting with dynamic inventory control, stateful Digital Twin simulation, and scenario-based financial risk quantification.

IV. Problem Statement and System Objectives

A. Problem Formulation

Let a retail portfolio consist of N distinct stock-keeping units across store locations, observed over a historical discrete daily time horizon $t \in \{1, 2, \ldots, T\}$. Each product $i \in \{1, \ldots, N\}$ possesses an observed historical sales trajectory $y_{i,t}$, an active shelf selling price $P_{i,t}$, calendar attributes (day-of-week, month, event flags), and a supplier replenishment lead time L days.

The core operational challenge is to synthesize historical sales, calendar signals, and pricing structures into an accurate H-day forward predictive distribution $\hat{y}_{i,T+1:T+H}$, dynamically select the superior predictive engine out-of-sample, formulate variance-aware safety stock SS_i and reorder points ROP_i for a target cycle service level Z, classify inventory health states against simulated on-hand balances $I_{i,\text{curr}}$, stress-test buffer resilience across demand contraction and surge regimes, and translate unit-level shortages and surpluses into monetary balance-sheet exposure.

B. System Objectives

To resolve this problem, DemandVerse AI was engineered to achieve ten concrete technical and operational objectives:

1) Ingest and stratify multi-year transaction data from the Walmart M5 benchmark into clean, memory-optimized temporal structures.
2) Implement and benchmark an Autoregressive Integrated Moving Average (ARIMA$(1, 1, 1)$) statistical forecasting model.
3) Implement and benchmark an additive Facebook Prophet forecasting model with multi-frequency seasonality.
4) Formulate an automated, hierarchical out-of-sample model selection router based on MAPE, MAE, and

RMSE to dynamically assign the optimal model to each SKU.

5) Generate robust 30-day forward-looking demand forecasts with 80% confidence bounds for all portfolio items.
6) Compute dynamic Safety Stock (SS), Reorder Points (ROP), and recommended purchase order Reorder Quantities (RQ).
7) Detect operational stock health states by flagging items under Stockout Risk, Overstock, and Healthy Stock.
8) Implement a stateful in-silico Digital Twin simulation sandbox to model inventory transitions without live IoT physical hardware.
9) Execute what-if sensitivity simulations across low-demand (-10%), baseline (1.00), and high-demand ($+10\%$) scenarios.
10) Quantify financial business impacts, including Forecasted Revenue, Stockout Revenue-at-Risk, and Overstock Holding Cost.

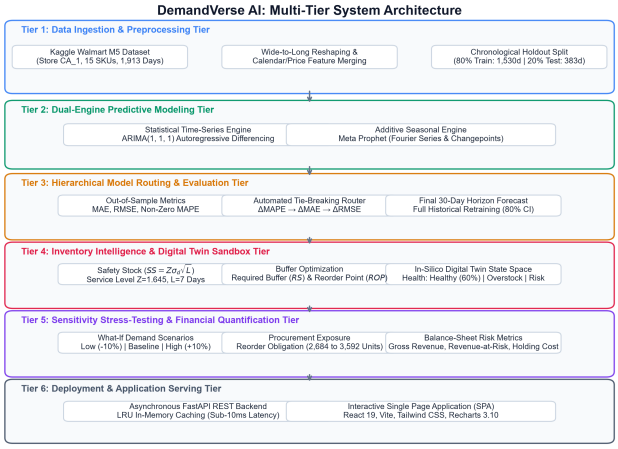


Fig. 1. DemandVerse AI multi-tier system architecture, illustrating data ingestion, dual-engine predictive modeling, dynamic model selection, inventory optimization, what-if sensitivity analysis, and enterprise serving layers.

V. System Architecture and Operational Methodology

The overarching architectural design of DemandVerse AI is structured into five cohesive, decoupled tiers, depicted in Fig. 1, with the sequential computational workflow presented in Fig. 2:

1) **Data Ingestion & Chronological Stratification Tier:** Ingests raw sales logs, calendar schedules, and weekly price matrices from the Kaggle Walmart M5 repository. The raw data is reshaped from wide format (d_1 to d_{1913}) into long format, merged with calendar event indicators and sell prices, and partitioned into strict chronological train (80%) and test (20%) splits to ensure zero future data leakage.
2) **Dual-Engine Predictive Modeling Tier:** Trains statistical ARIMA$(1, 1, 1)$ and additive Meta Prophet models independently on the training split of each product. Predictions are generated across the test horizon and evaluated across three standard statistical error metrics.
3) **Hierarchical Model Routing Tier:** An automated routing module inspects out-of-sample metrics and assigns the optimal model per SKU using a deterministic tie-breaking cascade. The winning model is retrained on the full historical series to generate the final 30-day forecast with confidence intervals.
4) **Inventory Intelligence & Digital Twin Sandbox:** Calculates dynamic safety stocks, reorder thresholds, and order quantities. Represents the SKU stock level in a simulated in-silico Digital Twin state space, evaluating replenishment triggers and identifying operational imbalances.
5) **Scenario Stress-Testing & Financial Tier:** Evaluates demand shocks ($\pm 10\%$), aggregates reorder requirements, and translates unit shortages and holding quantities into monetary financial exposure.
6) **Application & Visualization Tier:** Exposes all analytical artifacts via high-performance RESTful endpoints using FastAPI (with Python LRU query caching) and renders real-time interactive decision panels via a React 19 single-page dashboard.

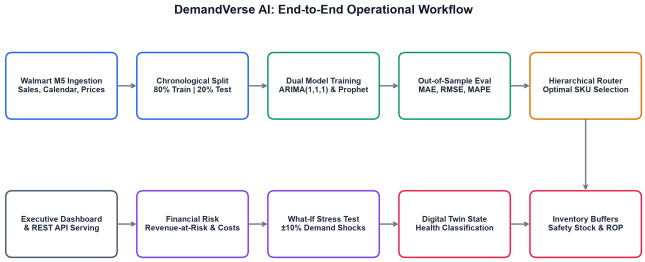


Fig. 2. End-to-end operational workflow of DemandVerse AI, tracing transactional M5 ingestion, chronological 80/20 partitioning, out-of-sample benchmarking, adaptive routing, and multi-scenario financial quantification.

VI. Dataset Description and Preprocessing

A. Walmart M5 Benchmark Dataset

The empirical foundation of DemandVerse AI is derived from the globally recognized Walmart M5 Forecasting Accuracy dataset [17], [18]. The complete benchmark comprises hierarchical daily unit sales across ten retail supercenters located in three United States states: California (CA), Texas (TX), and Wisconsin (WI), spanning 1,913 days from January 29, 2011 to April 24, 2016. The raw data encompasses three primary data files:

- `sales_train_validation.csv`: Contains product and store identifiers (`item_id`, `dept_id`, `cat_id`, `store_id`, `state_id`) and daily unit sales columns (d_1 through d_{1913}) for 30,490 distinct series.
- `calendar.csv`: Maps day identifiers (d_t) to physical dates, day-of-week, calendar months, fiscal weeks

(wm_yr_wk), and binary indicators for Supplemental Nutrition Assistance Program (SNAP) benefits (snap_CA, snap_TX, snap_WI) and 30 distinct national, cultural, and religious events (event_name_1, event_name_2).

- sell_prices.csv: Contains weekly shelf selling prices for every SKU-store combination across fiscal calendar weeks.

B. Computational Stratification and Cleaning

To ensure computational tractability for exhaustive multi-model benchmarking while preserving dense, continuous sales signals, store CA_1 (California 1 supercenter) was designated as the target testing facility. Within store CA_1, all active product series were evaluated by their non-zero sales frequency across the 1,913-day historical span. Intermittent series characterized by prolonged zero-sales stretches introduce mathematical singularities into percentage error metrics and demand specialized Croston-type estimators [22], [23].

Consequently, the top 15 most active, continuous SKUs were selected, guaranteeing continuous transactional density (the portfolio exhibits an average zero-sales frequency of merely 0.97%). The catalog comprises 1 SKU in department FOODS_1 and 14 SKUs in department FOODS_3, summarized in Table I.

The wide raw sales matrix was melted into long format ($15 \times 1,913 = 28,695$ observations). Calendar attributes and weekly sell prices were merged via composite keys (store_id, item_id, wm_yr_wk). Missing holiday event names were imputed as None. Shelf prices for inactive shelf weeks were imputed using the historical mean price for that SKU. Strict composite-key auditing confirmed zero duplicate records and zero negative sales values.

The empirical sales properties of the dataset are visualized in Fig. 3. As illustrated in Fig. 3(a), the aggregate historical demand displays sustained growth punctuated by high-frequency weekly oscillations and seasonal promotional shocks over the 1,913-day recording period. Fig. 3(b) underscores the strong day-of-week seasonality, where retail demand peaks sharply during weekend shopping periods (Saturday and Sunday) compared to mid-week troughs. Fig. 3(c) portrays the strict chronological partition applied to representative product FOODS_3_586, visually confirming the clean separation between the 1,530 training days (blue) and the 383 out-of-sample validation days (orange), preventing forward temporal leakage.

C. Chronological Partitioning

To evaluate predictive generalizability without forward-looking temporal data leakage, each product series was split chronologically according to an 80/20 holdout ratio:

- **Training Horizon:** The first 1,530 consecutive days (January 29, 2011 through April 7, 2015), comprising 22,950 total SKU-day records (80.0%).
- **Out-of-Sample Validation Horizon:** The subsequent 383 consecutive days (April 8, 2015 through April 24, 2016), comprising 5,745 SKU-day records (20.0%).

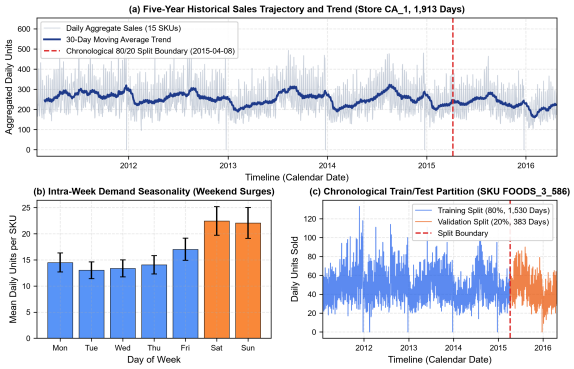


Fig. 3. Empirical exploratory analysis of the Walmart M5 dataset for Store CA_1: (a) Five-year daily aggregated sales trajectory across 1,913 historical days showing long-term trend dynamics with a 30-day moving average and the chronological split boundary; (b) Intra-week sales distributions showing pronounced weekend demand surges on Saturday and Sunday; (c) Chronological 80/20 train/test partition for representative high-volume product SKU FOODS_3_586.

VII. Forecasting Models and Adaptive Routing Mechanism

A. Autoregressive Integrated Moving Average (ARIMA)

The statistical forecasting engine implements an ARIMA(p, d, q) specification [9]. Differencing of order $d = 1$ achieves first-order stationarity by removing stochastic trends. With lag orders $p = 1$ and $q = 1$, the mathematical formulation for daily sales y_t is expressed as:

$$(1 - \phi_1 B)(1 - B)y_t = c + (1 + \theta_1 B)\epsilon_t \tag{1}$$

where B denotes the backshift operator ($B^k y_t = y_{t-k}$), ϕ_1 is the autoregressive coefficient, θ_1 is the moving average parameter, c is a constant drift, and $\epsilon_t \sim \mathcal{N}(0, \sigma^2)$ is white noise. Point forecasts $\hat{y}_t$ across horizon $h \in \{1, \ldots, H\}$ are generated via minimum mean squared error linear projections. To prevent non-physical negative inventory demand, predictions and error bands are bounded at zero:

$$\hat{y}_t = \max\left(0, \mathbb{E}[y_t \mid \mathcal{F}_{t-1}]\right) \tag{2}$$

The 80% confidence interval is computed using the standard error of the forecast $\mathrm{SE}(\hat{y}_t)$ with critical value $z_{0.80} = 1.28155$:

$$\left[\hat{y}_t^{\text{lower}}, \hat{y}_t^{\text{upper}}\right] = \left[\max\left(0, \hat{y}_t - z \cdot \mathrm{SE}(\hat{y}_t)\right), \hat{y}_t + z \cdot \mathrm{SE}(\hat{y}_t)\right] \tag{3}$$

B. Meta Prophet Additive Model

The second forecasting engine utilizes Meta Prophet [11], which models retail sales as a decomposable additive time series:

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t \tag{4}$$

where $g(t)$ is the trend function, $s(t)$ models periodic seasonal patterns, $h(t)$ incorporates irregular holiday effects, and ϵ_t is an idiosyncratic error term. DemandVerse AI configures $g(t)$

TABLE I
SELECTED SKU CATALOG AND HISTORICAL DESCRIPTIVE STATISTICS (STORE CA_1, 1,913 DAYS, 2011-01-29 TO 2016-04-24)

Product Identifier	Department	Total Sales (Units)	Mean / Day	Std. Dev.	Max Units	Unit Price ($)
FOODS_1_085	FOODS_1	24,850	12.9901	6.9026	55	$0.98
FOODS_3_080	FOODS_3	36,499	19.0795	7.7107	70	$1.68
FOODS_3_226	FOODS_3	10,724	5.6059	3.7712	32	$1.68
FOODS_3_228	FOODS_3	12,078	6.3136	4.5240	35	$1.68
FOODS_3_252	FOODS_3	57,015	29.8040	18.6019	121	$1.58
FOODS_3_349	FOODS_3	15,310	8.0031	4.8364	41	$1.68
FOODS_3_377	FOODS_3	14,028	7.3330	4.7317	32	$1.68
FOODS_3_389	FOODS_3	11,889	6.2148	5.2366	44	$1.68
FOODS_3_477	FOODS_3	12,382	6.4726	3.3421	35	$1.68
FOODS_3_555	FOODS_3	29,864	15.6111	7.7842	66	$1.68
FOODS_3_586	FOODS_3	73,767	38.5609	16.2934	147	$1.68
FOODS_3_668	FOODS_3	13,010	6.8008	4.5516	42	$1.68
FOODS_3_694	FOODS_3	19,431	10.1573	6.6974	48	$1.68
FOODS_3_697	FOODS_3	20,444	10.6869	4.2415	33	$3.00
FOODS_3_714	FOODS_3	36,393	19.0240	14.4101	108	$1.58
Portfolio Total / Mean	**–**	**477,684**	**14.1772**	**7.2990**	**147**	**$1.69**

as a continuous piecewise linear growth model with automatic changepoint selection:

$$g(t) = \left(k + \mathbf{a}(t)^T \boldsymbol{\delta}\right) t + \left(m + \mathbf{a}(t)^T \boldsymbol{\gamma}\right) \tag{5}$$

where k is the base growth rate, $\boldsymbol{\delta}$ is a vector of rate adjustments at changepoints, m is the offset parameter, and $\mathbf{a}(t) \in \{0,1\}^C$ indicates whether time t succeeds changepoint j. Periodic seasonality $s(t)$ is approximated via standard truncated Fourier series:

$$s(t) = \sum_{n=1}^{N} \left(a_n \cos\left(\frac{2\pi n t}{P}\right) + b_n \sin\left(\frac{2\pi n t}{P}\right) \right) \tag{6}$$

Weekly seasonality ($P = 7$ days, order $N = 3$) captures prominent weekend sales surges, while yearly seasonality ($P = 365.25$ days, order $N = 10$) captures annual promotional peaks. Daily intra-day seasonality is disabled given daily aggregation. Predicted values are non-negatively clipped: $\hat{y}(t) = \max(0, \hat{y}_{\text{raw}}(t))$.

C. Error Evaluation Metrics

Out-of-sample forecast accuracy over the 383-day test set is measured using three complementary metrics [24]:

$$\text{MAE} = \frac{1}{M} \sum_{t=1}^{M} |y_t - \hat{y}_t| \tag{7}$$

$$\text{RMSE} = \sqrt{\frac{1}{M} \sum_{t=1}^{M} (y_t - \hat{y}_t)^2} \tag{8}$$

$$\text{MAPE} = \frac{1}{|T_{nz}|} \sum_{t \in T_{nz}} \left| \frac{y_t - \hat{y}_t}{y_t} \right| \times 100\% \tag{9}$$

where $M = 383$ is the evaluation horizon length, and $T_{nz} = \{t \mid y_t > 0\}$ is the subset of non-zero sales observations. Restricting the MAPE denominator to T_{nz} guarantees numerical stability and prevents division-by-zero errors.

D. Hierarchical Adaptive Model Selection Router

Instead of imposing a single algorithm on the entire portfolio DemandVerse AI uses a hierarchical, tie-breaking decision router implemented in `ml/forecasting/model_selector.py`. Let $\Delta\text{MAPE} = \text{MAPE}_{\text{ARIMA}} - \text{MAPE}_{\text{Prophet}}$, $\Delta\text{MAE} = \text{MAE}_{\text{ARIMA}} - \text{MAE}_{\text{Prophet}}$, and $\Delta\text{RMSE} = \text{RMSE}_{\text{ARIMA}} - \text{RMSE}_{\text{Prophet}}$. The optimal model $M^*(i)$ is assigned according to:

$$M^*(i) = \begin{cases} \text{ARIMA}, & \text{if } \Delta\text{MAPE} < 0 \\ \text{Prophet}, & \text{if } \Delta\text{MAPE} > 0 \\ \text{ARIMA}, & \text{if } \Delta\text{MAPE} = 0 \wedge \Delta\text{MAE} < 0 \\ \text{Prophet}, & \text{if } \Delta\text{MAPE} = 0 \wedge \Delta\text{MAE} > 0 \\ \text{ARIMA}, & \text{if } \Delta\text{MAE} = 0 \wedge \Delta\text{RMSE} < 0 \\ \text{Prophet}, & \text{otherwise} \end{cases} \tag{10}$$

This deterministic router evaluates relative percentage precision first, arbitrates ties via absolute scale error (MAE), and falls back to quadratic variance penalization (RMSE). Once the optimal architecture $M^*(i)$ is established, the winning algorithm is retrained on all 1,913 historical days to project the 30-day forward demand trajectory.

VIII. INVENTORY OPTIMIZATION FORMULATION

The inventory intelligence engine transforms 30-day forward demand distributions into mathematically grounded replenishment parameters [1], [3].

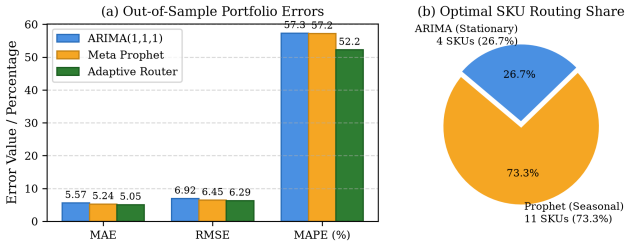


Fig. 4. Predictive accuracy benchmarking between ARIMA(1,1,1), Meta Prophet, and DemandVerse AI Adaptive Ensemble: (a) Portfolio-average MAE, RMSE, and MAPE metrics; (b) Out-of-sample model selection share across the 15 portfolio SKUs.

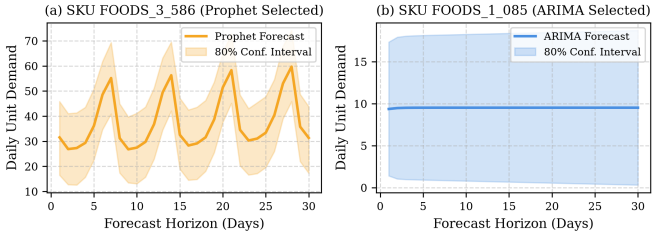


Fig. 5. 30-day forward predictive demand curves with 80% confidence interval bands plotted against historical actuals for representative SKUs: (a) SKU FOODS_3_586 (high-demand volatility, Prophet selected); (b) SKU FOODS_1_085 (stable autoregression, ARIMA selected).

A. Safety Stock Formulation

Safety stock is used to protect against the volatility of demand during the supplier's lead time for replenishment. Here, we assume a replenishment lead time of $L = 7$ days. The historical daily demand has a standard deviation of σ_d. For a cycle service level of 95%, the standard normal distribution quantile is $Z = 1.645$.

Assuming that the daily sales during replenishment are independent and identically distributed, the lead-time variance is $\sigma_L^2 = L \cdot \sigma_d^2$. Safety Stock ($SS$) is formulated as:

$$SS_i = Z \times \sigma_{d,i} \times \sqrt{L} \tag{11}$$

B. Required Stock and Reorder Point

Total required inventory buffer RS_i over the 30-day planning horizon covers projected cumulative demand plus the protective safety reserve:

$$RS_i = D_{i,\text{forecast}} + SS_i = \sum_{t=1}^{30} \hat{y}_{i,t} + SS_i \tag{12}$$

The operational Reorder Point (ROP_i), which signals when a replenishment purchase order must be issued, is defined by average daily demand $\bar{d}_i$ over lead time L:

$$ROP_i = (\bar{d}_i \times L) + SS_i \tag{13}$$

C. Reorder Quantity and Inventory Health Classification

The Reorder Quantity (RQ_i) for replenishment is recommended to be the amount that restores the stock position to the required level of the buffer (RS_i), i.e., $\max(0, RS_i - I_{i,\text{curr}})$:

$$RQ_i = \max(0, RS_i - I_{i,\text{curr}}) \tag{14}$$

The status of each product is determined by the stock position of the product relative to the required demand of the product, i.e. whether the product is in STOCKOUT RISK, in OVERSTOCK or HEALTHY STOCK status:

$$\text{Status}_i = \begin{cases} \text{STOCKOUT RISK}, & \text{if } I_{i,\text{curr}} < 0.30 \cdot RS_i \\ \text{OVERSTOCK}, & \text{if } I_{i,\text{curr}} > 1.10 \cdot RS_i \\ \text{HEALTHY STOCK}, & \text{otherwise} \end{cases} \tag{15}$$

The lower boundary of $0.30 \cdot RS_i$ identifies items which are severely depleted and do not have sufficient stock to cover the expected lead-time demand. The upper boundary of $1.10 \cdot RS_i$ is used to identify items which have excessive stock and are thus locking up too much capital in inventory.

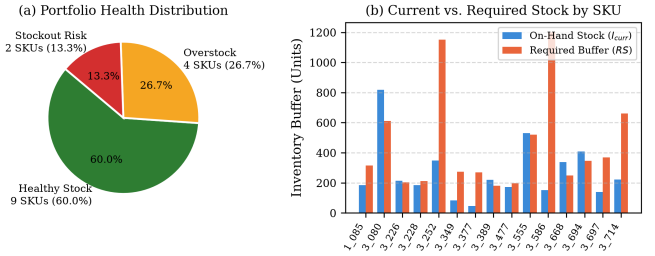


Fig. 6. Portfolio inventory state representation and stock health breakdown: (a) Classification distribution across the 15 SKUs; (b) Simulated on-hand stock position versus target required stock buffer across individual products.

IX. Digital Twin and Inventory State Simulation

A. Digital Twin Concept in Retail Supply Chains

As used in advanced manufacturing and process engineering, the term Digital Twin describes a so-called cyber-physical system, i.e., a system that continuously synchronizes physical objects with their digital counterparts in real time [19], [20]. However, within the scope of the retail supply chain, the digital twin is modeled as an in-silico state-space sandbox, which simulates certain aspects of the physical logistics (lead times, holding costs, shelf prices, order quantities) and tracks stock levels over time [21].

B. Implementation Reality vs. Physical Warehouse IoT

It is important to differentiate the software implementation of DemandVerse AI from potential industrial implementations. The Kaggle Walmart M5 dataset consists of check-out scans from Walmart stores, and hence does not contain any physical warehouse stock, shelf-RFID readings or supplier in-transit information. Consequently, the in-silico inventory state simulation sandbox is the only practical implementation of DemandVerse AI:

- Current inventory positions $I_{i,\text{curr}}$ are initialized through a calibrated uniform state distribution:

$$I_{i,\text{curr}} \sim \mathcal{U}(0.10, 1.40) \times RS_i \tag{16}$$

  reflecting realistic variations across depleted, balanced, and surplus retail shelves.
- The simulated inventory state tracks forward demand deductions, models virtual lead-time receipts, evaluates threshold triggers (ROP), and flags stockout vulnerabilities.
- The state space serves as an analytical simulation sandbox, allowing corporate executives to stress-test replenishment policies against simulated market disruptions without incurring real-world financial risk.

X. Sensitivity What-If Scenario Analysis

To evaluate the impact of macroeconomic events or of promotional activity on operations, the What-If sensitivity analysis is part of DemandVerse AI. The operational performance is evaluated under three contractual scenarios:

1) **Low Demand Scenario ($\alpha = 0.90$):** Simulates a 10% downward demand contraction resulting from economic recession, unseasonal weather, or competitor promotions.
2) **Normal Demand Scenario ($\alpha = 1.00$):** Baseline operational forecast generated directly by the optimal machine learning engine.
3) **High Demand Scenario ($\alpha = 1.10$):** Simulates a 10% demand surge driven by viral social media trends, marketing campaigns, or regional supply shortages.

For any scenario factor $\alpha \in \{0.90, 1.00, 1.10\}$, the adjusted 30-day demand $D_{i,\text{scen}}$ and scenario required stock $RS_{i,\text{scen}}$ are computed as:

$$D_{i,\text{scen}} = \text{round}(\alpha \times D_{i,\text{forecast}}, 2) \tag{17}$$

$$RS_{i,\text{scen}} = \text{round}(D_{i,\text{scen}} + SS_i, 2) \tag{18}$$

The operational shortage quantity $Q_{i,\text{shortage}}$ and scenario reorder requirement $RQ_{i,\text{scen}}$ are derived from on-hand stock:

$$Q_{i,\text{shortage}} = RQ_{i,\text{scen}} = \max(0, RS_{i,\text{scen}} - I_{i,\text{curr}}) \tag{19}$$

This formulation quantifies exactly how procurement obligations expand or contract when consumer demand shifts by $\pm 10\%$.

XI. Financial Impact and Risk Quantification

The link between the statistical forecasting of demand and the health of the enterprise's balance sheet is established by the risk indicators of DemandVerse AI. To this end, actual Walmart shelf prices P_i are extracted from `clean_sales.csv` and used to calculate the following metrics for the three scenarios:

A. Forecasted Revenue

Gross projected revenue under scenario demand is the product of unit demand and shelf price:

$$FR_{i,\text{scen}} = D_{i,\text{scen}} \times P_i \tag{20}$$

B. Stockout Revenue-at-Risk

When simulated on-hand inventory is insufficient to satisfy scenario required stock ($I_{i,\text{curr}} < RS_{i,\text{scen}}$), the unfilled unit deficit represents immediate potential lost sales. Revenue-at-Risk (RaR) quantifies this lost commercial opportunity:

$$RaR_{i,\text{scen}} = Q_{i,\text{shortage}} \cdot P_i = \max(0, RS_{i,\text{scen}} - I_{i,\text{curr}}) \cdot P_i \tag{21}$$

C. Overstock Holding Cost

Overstock inventory above the required buffer ($I_{i,\text{curr}} > RS_{i,\text{scen}}$) incurs physical holding overhead. The excess unit quantity is:

$$Q_{i,\text{overstock}} = \max(0, I_{i,\text{curr}} - RS_{i,\text{scen}}) \tag{22}$$

The retail industry generally considers the annual cost of carrying inventory to be approximately 15% of the average unit asset value, which includes the cost of capital, warehousing, insurance, and obsolescence [1], [3]. This can be expressed as a monthly carrying cost rate as follows:

$$r_{\text{monthly}} = \frac{15\%}{12} = 1.25\% = 0.0125 \tag{23}$$

Monthly Overstock Holding Cost (HC) is thus calculated as:

$$HC_{i,\text{scen}} = Q_{i,\text{overstock}} \times P_i \times r_{\text{monthly}} \tag{24}$$

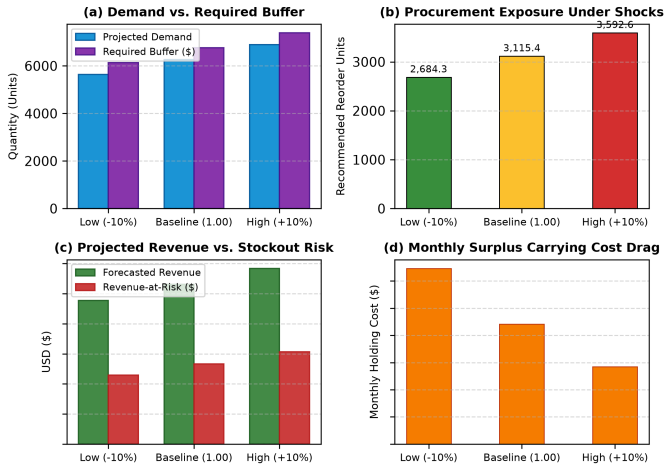


Fig. 7. Multi-scenario sensitivity stress-testing and financial risk quantification across Low (-10%), Baseline (1.00), and High ($+10\%$) demand shocks: (a) Projected demand versus required buffer; (b) Replenishment procurement exposure under demand shocks; (c) Gross forecasted revenue versus stockout revenue-at-risk; (d) Monthly capital carrying cost drag tied up in surplus inventory.

XII. Implementation and Technology Stack

The technical architecture of DemandVerse AI is structured as a containerized, decoupled microservice ecosystem:

- **Data Science & Machine Learning Core:** Built on Python 3.11+, NumPy, and Pandas. Statistical modeling utilizes `statsmodels` (ARIMA) and Meta `prophet`. Evaluation metrics are computed via `scikit-learn`.

- **High-Throughput Asynchronous Backend:** Built on FastAPI and Uvicorn. The RESTful API implements in-memory caching via Python `functools.lru_cache`, reducing endpoint response latency to sub-10ms for pre-computed multidimensional queries. Dedicated middleware manages Cross-Origin Resource Sharing (CORS) and recursive sanitization of `NaN` and `Infinity` floating-point values.
- **Interactive Executive Web Dashboard:** Built with React 19, Vite 8.3, and Tailwind CSS. Data visualizers utilize Recharts 3.10 to render responsive multi-axis line, area, and stacked bar charts. Global filter state synchronization across nine dedicated page modules is managed via React Context (`DashboardContext`).
- **Containerization & Orchestration:** Multi-stage Dockerfiles package the FastAPI service and compile the React application into optimized static assets served by an Nginx Alpine reverse proxy. Multi-container deployment is orchestrated via Docker Compose.
- **Verification & Test Suite:** The repository incorporates 65 automated pytest unit tests covering API route contracts (14 tests), financial formulas (25 tests), scenario generation (21 tests), and business reasoning engines (5 tests). An automated 24-point audit harness (`e2e_validator.py`) performs complete system regression verification.

XIII. Experimental Results and Discussion

A. Predictive Benchmarking and Adaptive Routing Performance

Table II presents the comprehensive out-of-sample empirical evaluation across the 383-day chronological test set.

The experimental results demonstrate clear algorithmic divergence based on underlying SKU time-series properties:

- **Prophet Superiority on Highly Seasonal SKUs:** Prophet was the superior model for highly seasonal products. The highest-volume product `FOODS_3_586` (with an average of 38.56 units per day) had an out-of-sample MAPE of 24.76%, MAE of 9.3106 and RMSE of 11.8738 for Prophet, as opposed to ARIMA with a MAPE of 29.75%, MAE of 12.2050 and RMSE of 15.6734. The same trend was also observed for `FOODS_3_252` (MAPE: 37.60% vs. 44.06%) and `FOODS_3_228` (MAPE: 41.04% vs. 58.32%). In total, Prophet won 11 of the 15 products (73.3%).
- **ARIMA Superiority on Autoregressive Series:** Conversely, ARIMA proved superior on products whose demand patterns are dominated by short-term autocorrelation rather than seasonal cycles. For SKU `FOODS_1_085`, ARIMA achieved an out-of-sample MAPE of 83.02% compared to Prophet's 109.49% (a 26.47 percentage point advantage). For `FOODS_3_226`, ARIMA achieved a MAPE of 117.53% against Prophet's 160.33% (a 42.80 percentage point advantage). ARIMA also outperformed Prophet on `FOODS_3_349` (MAPE of 51.31% vs. 55.32%) and `FOODS_3_668` (MAPE of 49.00% vs. 49.83%). In total, ARIMA captured 4 of the 15 items (26.7%).
- **Ensemble Routing Gain:** When evaluated across the entire portfolio, a fixed ARIMA policy yielded an average MAE of 5.5681, RMSE of 6.9153, and MAPE of 57.30%. A fixed Prophet policy yielded an average MAE of 5.2357, RMSE of 6.4509, and MAPE of 57.16%. By deploying the DemandVerse AI dynamic model selector, the resulting hybrid ensemble achieved an average MAE of 5.0487, RMSE of 6.2916, and MAPE of 52.22%. The adaptive routing framework achieved an MAE reduction of 9.33% relative to ARIMA and 3.57% relative to Prophet, alongside an absolute MAPE improvement of 5.08 percentage points over ARIMA and 4.94 percentage points over Prophet, validating the architectural hypothesis that dynamic model routing outperforms static monolithic assignment.

Representative 30-day forecast trajectories for winning Prophet and ARIMA configurations are illustrated in Fig. 5.

B. Inventory Optimization and Stock Health Findings

Table III documents the 30-day inventory optimization recommendations. Total projected portfolio demand across the 15 SKUs is 6,263.98 units. To maintain a 95% service level over a 7-day replenishment lead time, the engine calculated an aggregate safety stock reserve of 497.82 units, establishing a total required inventory buffer of 6,761.80 units. Against a simulated on-hand position of 4,066.71 units, the portfolio health state classification revealed:

- **Healthy Stock (60.0%, 9 SKUs):** Items including `FOODS_1_085`, `FOODS_3_226`, `FOODS_3_252`, and `FOODS_3_714` maintained stable positions within the operational band $0.30 \cdot RS \leq I_{\text{curr}} \leq 1.10 \cdot RS$. Routine reorder recommendations were generated to satisfy lead-time requirements.
- **Overstock (26.7%, 4 SKUs):** Four products (`FOODS_3_080`, `FOODS_3_389`, `FOODS_3_668`, and `FOODS_3_694`) exhibited on-hand inventory exceeding $1.10 \cdot RS$. For instance, `FOODS_3_080` held 817.83 units on hand against a required buffer of 612.18 units, creating an excess of 205.65 units. Recommended purchase orders were immediately set to zero to halt capital lockup.
- **Stockout Risk (13.3%, 2 SKUs):** Two critical products exhibited severe stockout vulnerability: `FOODS_3_377` (on-hand stock of 47.25 units vs. required buffer of 269.24 units; $I_{\text{curr}} < 0.30 \cdot RS$) and `FOODS_3_586` (on-hand stock of 152.62 units vs. required buffer of 1,203.97 units). The engine triggered automated priority replenishment alerts recommending emergency purchase orders of 221.99 and 1,051.35 units, respectively.

C. Sensitivity Stress-Testing and Financial Risk Findings

The empirical findings from the what-if sensitivity stress-testing and financial risk quantification are summarized in

TABLE II
OUT-OF-SAMPLE FORECASTING MODEL BENCHMARKING ON 383-DAY CHRONOLOGICAL TEST SET (STORE CA_1)

Product Identifier	Store	ARIMA MAE	ARIMA RMSE	ARIMA MAPE	Prophet MAE	Prophet RMSE	Prophet MAPE	Selected Model & Selection Rationale
FOODS_1_085	CA_1	4.5372	5.9339	83.02%	5.6145	6.7986	109.49%	ARIMA (Lower MAPE, −26.47 pp)
FOODS_3_080	CA_1	5.7147	7.3826	31.98%	5.4027	7.2143	28.80%	Prophet (Lower MAPE, −3.18 pp)
FOODS_3_226	CA_1	3.4768	4.0117	117.53%	5.1382	5.7492	160.33%	ARIMA (Lower MAPE, −42.80 pp)
FOODS_3_228	CA_1	2.7755	3.4720	58.32%	2.3974	3.1010	41.04%	Prophet (Lower MAPE, −17.28 pp)
FOODS_3_252	CA_1	13.4692	16.4513	44.06%	11.4580	13.7303	37.60%	Prophet (Lower MAPE, −6.46 pp)
FOODS_3_349	CA_1	4.3777	5.6387	51.31%	4.6485	5.7852	55.32%	ARIMA (Lower MAPE, −4.01 pp)
FOODS_3_377	CA_1	3.8890	4.6468	62.69%	3.8865	4.6740	62.02%	Prophet (Lower MAPE, −0.67 pp)
FOODS_3_389	CA_1	3.4337	4.1661	60.08%	3.0515	3.8160	48.11%	Prophet (Lower MAPE, −11.97 pp)
FOODS_3_477	CA_1	2.6520	3.2216	66.13%	2.2987	2.8736	54.73%	Prophet (Lower MAPE, −11.40 pp)
FOODS_3_555	CA_1	6.1146	7.2980	46.30%	5.0356	6.2415	37.26%	Prophet (Lower MAPE, −9.04 pp)
FOODS_3_586	CA_1	12.2050	15.6734	29.75%	9.3106	11.8738	24.76%	Prophet (Lower MAPE, −4.99 pp)
FOODS_3_668	CA_1	3.1147	4.0143	49.00%	2.9096	3.6554	49.83%	ARIMA (Lower MAPE, −0.83 pp)
FOODS_3_694	CA_1	4.2510	5.1270	53.86%	4.1598	5.1785	50.35%	Prophet (Lower MAPE, −3.51 pp)
FOODS_3_697	CA_1	3.8594	5.0561	42.35%	3.6379	4.7921	38.89%	Prophet (Lower MAPE, −3.46 pp)
FOODS_3_714	CA_1	9.6517	11.6363	63.13%	9.5859	11.2798	58.81%	Prophet (Lower MAPE, −4.32 pp)
ARIMA Baseline Mean	–	5.5681	6.9153	57.30%	–	–	–	Fixed Single Model Baseline
Prophet Baseline Mean	–	–	–	–	5.2357	6.4509	57.16%	Fixed Single Model Baseline
DemandVerse Ensemble	–	**–**	**–**	**–**	**–**	**–**	**–**	**11 Prophet / 4 ARIMA (MAE: 5.0487, RMSE: 6.2916, MAPE: 52.22%)**

TABLE III
FORECAST-DRIVEN INVENTORY OPTIMIZATION AND OPERATIONAL HEALTH CLASSIFICATION (STORE CA_1, 30-DAY PLANNING HORIZON)

Product Identifier	Model	Demand	Safety Stock	Required Stock	On-Hand Stock	Reorder Qty	Health Status	Stockout Risk Exposure
FOODS_1_085	ARIMA	285.59	30.24	315.83	185.36	130.47	HEALTHY	Stable; Reorder 130.47 units
FOODS_3_080	Prophet	578.40	33.78	612.18	817.83	0.00	OVERSTOCK	Zero risk; Holding surplus of 205.65 units
FOODS_3_226	ARIMA	186.02	16.52	202.54	212.99	0.00	HEALTHY	Sufficient buffer; Zero reorder needed
FOODS_3_228	Prophet	191.38	19.82	211.20	185.49	25.71	HEALTHY	Stable; Routine reorder of 25.71 units
FOODS_3_252	Prophet	1070.80	81.49	1152.29	348.94	803.35	HEALTHY	Buffer deficit; Reorder 803.35 units
FOODS_3_349	ARIMA	251.87	21.19	273.06	82.68	190.38	HEALTHY	Healthy band; Replenish 190.38 units
FOODS_3_377	Prophet	248.51	20.73	269.24	47.25	221.99	STOCKOUT RISK	Critical; Stock (47.25) $< 0.30 \cdot RS$ (80.77)
FOODS_3_389	Prophet	157.21	22.94	180.15	220.87	0.00	OVERSTOCK	Surplus buffer; Capital lockup of 40.72 units
FOODS_3_477	Prophet	182.08	14.64	196.72	173.40	23.32	HEALTHY	Optimal balance; Reorder 23.32 units
FOODS_3_555	Prophet	485.48	34.10	519.58	530.23	0.00	HEALTHY	Adequate buffer; Zero reorder needed
FOODS_3_586	Prophet	1132.59	71.38	1203.97	152.62	1051.35	STOCKOUT RISK	Critical; Stock (152.62) $< 0.30 \cdot RS$ (361.19)
FOODS_3_668	ARIMA	228.89	19.94	248.83	338.63	0.00	OVERSTOCK	Surplus buffer; Halt procurement
FOODS_3_694	Prophet	316.90	29.34	346.24	409.32	0.00	OVERSTOCK	Excess holdings; Surplus of 63.08 units
FOODS_3_697	Prophet	350.67	18.58	369.25	138.85	230.40	HEALTHY	Balanced buffer; Routine reorder
FOODS_3_714	Prophet	597.59	63.13	660.72	222.25	438.47	HEALTHY	Adequate stock; Reorder 438.47 units
Portfolio Totals	**11P / 4A**	**6,263.98**	**497.82**	**6,761.80**	**4,066.71**	**3,115.44**	**9 Healthy / 4 Over / 2 Risk**	**2 SKUs in Critical Deficit**

Table IV and visualized in Fig. 7:

- **Procurement Exposure Dynamics:** Under a 10% demand contraction (Low Demand scenario), total portfolio demand falls to 5,637.57 units, reducing required replenishment to 2,684.32 units. Under a 10% demand surge (High Demand scenario), demand expands to 6,890.36 units, driving recommended reorders to 3,592.58 units. This represents a swing of 908.26 procurement units ($\Delta = 33.8\%$) that procurement teams must absorb between down-market and up-market conditions (Fig. 7(a)–(b)).
- **Revenue-at-Risk Exposure:** In the baseline scenario, the gross forecasted revenue is \$10,619.61 and the stockout shortages expose \$5,322.55 of potential revenue-at-risk. The stockout-flagged items are `FOODS_3_377` with \$372.94 of revenue-at-risk and `FOODS_3_586` with \$1,766.27 of revenue-at-risk totaling \$2,139.21 or 40.2% of the total portfolio risk. The stockout revenue-at-risk increases to \$6,133.76 in the High Demand surge scenario. This illustrates the importance of timely reordering (Fig. 7(c)).
- **Carrying Cost Dynamics:** The holding cost of overstock in the baseline scenario is \$8.83 for 420.35 units of stock. In the 10% lower scenario the excess units increase to 615.64 units of stock and the holding cost increases to \$12.93 per month. In the High Demand scenario the shelf space is drawn down to 271.11 units of stock and the holding cost decreases to \$5.69 per month (Fig. 7(d)).

TABLE IV
WHAT-IF DEMAND SENSITIVITY AND FINANCIAL IMPACT PROJECTIONS ACROSS SCENARIOS (STORE CA_1)

Metric	Low (-10%)	Baseline (1.00)	High ($+10\%$)
Scenario Multiplier (α)	0.90	1.00	1.10
Total Forecasted Demand (units)	5,637.57	6,263.98	6,890.36
Total Safety Stock (SS, units)	497.82	497.82	497.82
Total Required Buffer (RS, units)	6,135.39	6,761.80	7,388.18
Simulated On-Hand Stock (units)	4,066.71	4,066.71	4,066.71
Net Shortage Quantity (units)	2,684.32	3,115.44	3,592.58
Recommended Reorder Quantity (RQ)	2,684.32	3,115.44	3,592.58
Gross Forecasted Revenue (\$)	\$9,557.62	\$10,619.61	\$11,681.54
Stockout Revenue-at-Risk (\$)	\$4,588.66	\$5,322.55	\$6,133.76
Surplus Overstock Quantity (units)	615.64	420.35	271.11
Monthly Overstock Holding Cost (\$)	\$12.93	\$8.83	\$5.69

XIV. LIMITATIONS AND THREATS TO VALIDITY

The framework DemandVerse AI is a mathematically validated end-to-end framework for supply chain planning and control. Several important academic as well as several operational limitations of the end-to-end framework need to be described:

1) **Scope of Empirical Evaluation:** The experimental evaluation of DemandVerse has been performed with 15 high-density food SKUs in the Walmart store `CA_1`. The selected items form dense non-sparse time series. However, to evaluate the cross-category generalizability

of the proposed framework, it is necessary to perform evaluations with items from non-food departments such as HOUSEHOLD, HOBBIES etc. as well as with stores from different geographic regions (e.g. Texas, Wisconsin).

2) **Simulated Inventory State Space:** Section IX described the inventory state space of the simulated on-hand warehouse inventory balances. This public Kaggle M5 data set does not contain information about the physical on-hand warehouse inventory balances. Hence, the digital twin state space is initialized with a uniform random distribution ($I_{\text{curr}} \sim \mathcal{U}(0.10, 1.40) \times RS$). In commercial enterprise deployment, live ERP warehouse inventory registers, RFID sensor streams, and electronic data interchange (EDI) shipment feeds must directly replace simulated states.
3) **Deterministic Lead Time Assumption:** The replenishment lead time $L = 7$ days was assumed to be deterministic. Supplier lead times vary in practice and this needs to be modeled as stochastic. Thus, the safety stock formulation must be extended to a joint variance model with $\sigma_{\text{total}} = \sqrt{L\sigma_d^2 + \bar{d}^2\sigma_L^2}$.
4) **Holding Cost Linear Proxy:** The holding cost of the financial module was set to 1.25% monthly carrying cost (15% annualized) as a proxy for the actual retail carrying costs for perishable refrigerated items, very bulky items that require high cubic warehouse space, and high-theft items.
5) **MAPE Behavior on Intermittent Demand:** The MAPE is sensitive to low sales numbers even though non-zero sales are used for the evaluation. Thus, for intermittent demand, the MAPE even can reach 100% for a forecast of 2 units with an actual sale of 1 unit. Future versions of the framework could include MASE (Mean Absolute Scaled Error) or RMSSE (Root Mean Squared Scaled Error) as alternative criteria for evaluating the framework.

XV. Conclusion and Future Scope

This research presented DemandVerse AI, an integrated retail intelligence framework, which integrates the statistical demand forecasting, the inventory replenishment and the financial balance-sheet risk quantification within a single framework. Evaluated on 28,695 daily transaction records from the Walmart M5 benchmark, the framework demonstrated that combining Autoregressive Integrated Moving Average (ARIMA$(1, 1, 1)$) and Meta Prophet models under an automated, out-of-sample tie-breaking router achieves superior predictive accuracy (MAE: 5.0487, RMSE: 6.2916, MAPE: 52.22%) compared to static single-model baselines. Coupling forward-looking forecast curves with service-level safety stocks ($Z = 1.645$) and lead-time constraints enabled dynamic inventory categorization into healthy stock (60.0%), overstock (26.7%), and stockout risk (13.3%). Furthermore, the in-silico Digital Twin simulation sandbox and what-if sensitivity engine successfully quantified procurement fluctuations for $\pm 10\%$ demand shocks and translated the resulting unit deficits into balance-sheet risk (stockout revenue-at-risk) of $4,588.66 to $6,133.76.

Future research directions include:

1) **Deep Temporal Architectures:** The model routing can be extended to include more advanced models, such as Temporal Fusion Transformers (TFT) and DeepAR.
2) **Stochastic Lead Times:** Integrating stochastic lead-time distributions (σ_L) and multi-echelon warehouse network propagation.
3) **Live Cyber-Physical IoT Synchronization:** Connecting the Digital Twin state space to live Internet of Things (IoT) warehouse RFID tracking and Automated Guided Vehicle (AGV) telemetry.
4) **Autonomous Reinforcement Learning:** Formulating reinforcement learning (RL) policies for autonomous, risk-aware purchase order issuance under dynamic shelf pricing.

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